

Vintage Automation - Meeting with Roy

January 2026

One-Sentence Summary: Built a working Vintage Automation engine as a **pilot project** that demonstrates the SuperFact 4-layer framework - pulling experiment data from tactic tables, discovering channels automatically, and producing vintage curves with lift and confidence intervals.

The Story: Why This Matters

The problem we're solving:

- Analysts spend weeks recreating the same vintage curve logic for each campaign
- "Success" means different things to different people
- No systematic way to pull campaign context from data - everything is hardcoded

What Vintage Automation proves:

- We CAN pull experiment metadata directly from tactic tables (Layer 1)
- We CAN structure code so campaign/success definitions are "swap points" for future data sources
- We CAN discover things like **channel** from the data instead of hardcoding
- The 4-layer architecture actually works in practice

How Vintage Automation Fits SuperFact

VINTAGE AUTOMATION (PILOT)

Layer 1
Experiment
Metadata

```
+-----+  
| tactic_ |  
| evnt_hist |  
|         |
```

Layer 2
Campaign
Metadata

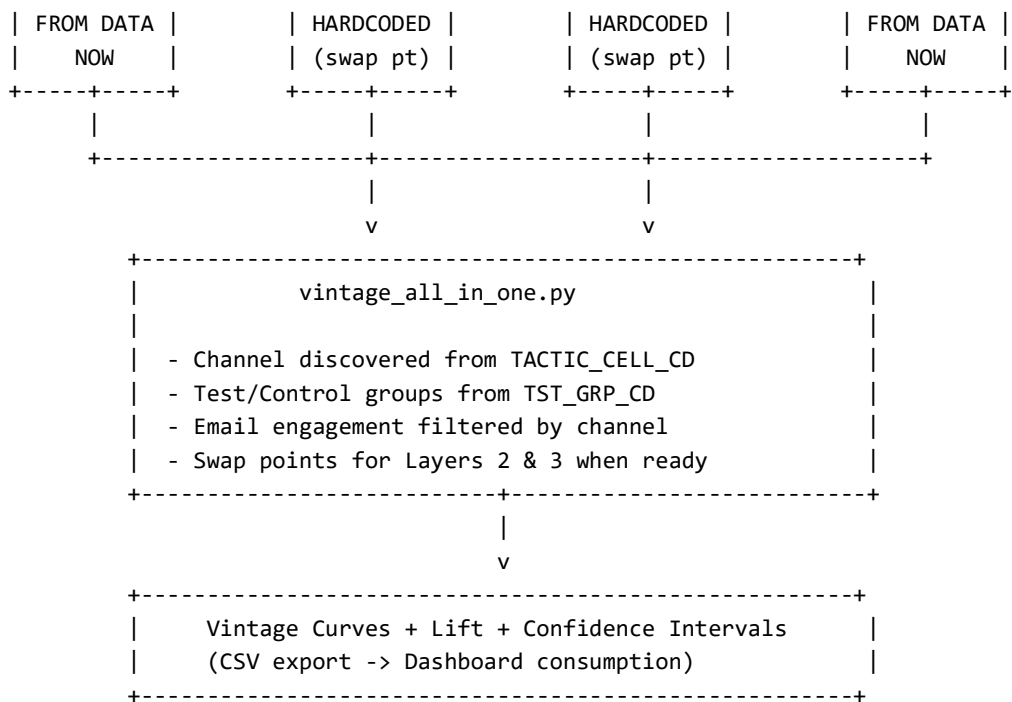
```
+-----+  
| Mnemonic |  
| Mapping  |  
| v2       |
```

Layer 3
Success
Library

```
+-----+  
| GitHub   |  
| Success  |  
| Logic    |
```

Layer 4
Client
Journey

```
+-----+  
| VISA_DR_ |  
| CRD       |  
| EMAIL     |
```



What's Built & What's Changed

Current State

Component	Status	Recent Progress
Layer 1: Experiment data (tactic)	FROM DATA	Correct path, partition pruning
Layer 2: Campaign metadata	Hardcoded	Swap point documented
Layer 3: Success definitions	Hardcoded	Swap point documented
Layer 4: Success outcome	FROM DATA	Card acquisition, activation, usage, tokenization
Layer 4: Email engagement	FROM DATA	Filtered by channel (only where channel contains "EM")
Channel detection	FROM DATA	Discovered from TACTIC_CELL_CD, handles combos like EM_IM

Vintage calculation	Working	COHORT + GROUP level, lift, CI
CSV/HDFS export	Working	Ready for dashboard consumption

Key Change: Channel Now From Data

Before (wrong):

```
# Hardcoded channel in config
CAMPAIGN_CONFIG = {
  "VCN": {
    "channel": "EMAIL", # <- Hardcoded
  }
}
```

After (correct):

```
# Discovered from data
channel_counts = tactic_df.groupby(F.trim(F.col("TACTIC_CELL_CD"))).count()
# Result: {'MB': 13M, 'XX': 762K, 'EM': 5K, 'EM_IM': 2K, ...}

# Email engagement only for clients where channel contains "EM"
email_clients = tactic_df.filter(F.trim(F.col("TACTIC_CELL_CD")).contains("EM"))
```

Why this matters:

- CONTROL group has no real channel (they receive no contact)
- Channel is a breakdown variable, not a grouping dimension for lift
- Combo channels like EM_IM (email + internet banking) are handled correctly

Campaigns Covered (6 Pilots)

MNE	Campaign	Success Type	Status
VCN	Contextual Notification	Acquisition	Ready to test
VDA	Black Friday Cyber Monday	Acquisition	Config exists
VDT	Activation Trigger	Activation	Config exists
VUI	Usage Trigger	Usage	Config exists
VUT	Tokenization Usage	Tokenization	Config exists
VAW	Add To Wallet	Tokenization	Config exists

The Philosophy: What to Hardcode vs Pull From Data

Data Source	Approach	Why
Experiment data (tactic_evnt_hist)	FROM DATA	Systematic access exists
Channel (TACTIC_CELL_CD)	FROM DATA	Systematic access exists
Test groups (TST_GRP_CD)	FROM DATA	Systematic access exists
Campaign metadata	HARDCODED (swap point)	Mnemonic Mapping v2 not ready
Success definitions	HARDCODED (swap point)	Success Library not ready

Swap points are documented - when the proper data sources exist, we replace the hardcoded dicts with queries.

What We Learned (Important for Future)

1. CONTROL Has No Channel

CONTROL clients receive no contact, so they have no meaningful channel (usually marked "XX" or similar). Vintage curves must compare TEST vs CONTROL at **COHORT level**, not by channel.

2. Channels Can Be Combos

A client targeted by both email and internet banking has channel `EM_IM`. Email engagement filtering uses `.contains("EM")` to catch all combos.

3. Data Quality Issues

- TACTIC_CELL_CD values have trailing spaces (need `F.trim()`)
- Some fields are consistently null (MDM fields)

4. Test Group Flexibility Still Needed

Currently hardcoded `TG4 = Test`. Some campaigns have A/B tests where TG1 and TG4 are both "action" groups. Need to make this configurable per campaign.

What's Still Needed

High Priority

Item	Status	Blocker
Validate VCN against known results	Not started	Need known baseline
Test group flexibility (A/B tests)	Not done	Need TST_GRP_CD meanings
Fulfillment code mapping	Not done	Need expert input
Metric dropdown in dashboard	Not done	Design ready (see below)

Medium Priority

Item	Status	Notes
Channel breakdown in dashboard	Function exists	Not integrated yet
RPT_GRP_CD segmentation	Field loaded	Don't know what it means

Questions for Experts

1. Where are fulfillment codes documented?
2. What are all TST_GRP_CD values and their meanings?
3. What does RPT_GRP_CD represent?

Key Feature: Metric Dropdown (Vintage Any Metric)

The Problem

Current: Email engagement shown as **overall summary** (one number for whole campaign).

Over 2 years of cohorts, overall summary is useless. Director needs: "Show me unsubscribe rate **by cohort**" - not "overall unsubscribe is 2%".

The Solution

Dropdown to select which metric to plot as a vintage curve:

```
+-----+
| Metric: [Primary Success v] |
| +- Primary Success (Acquisition) |
| +- Open Rate |
| +- Click Rate |
| +- Unsubscribe Rate |
| +- Bounce Rate |
+-----+
```

Same vintage curve structure:

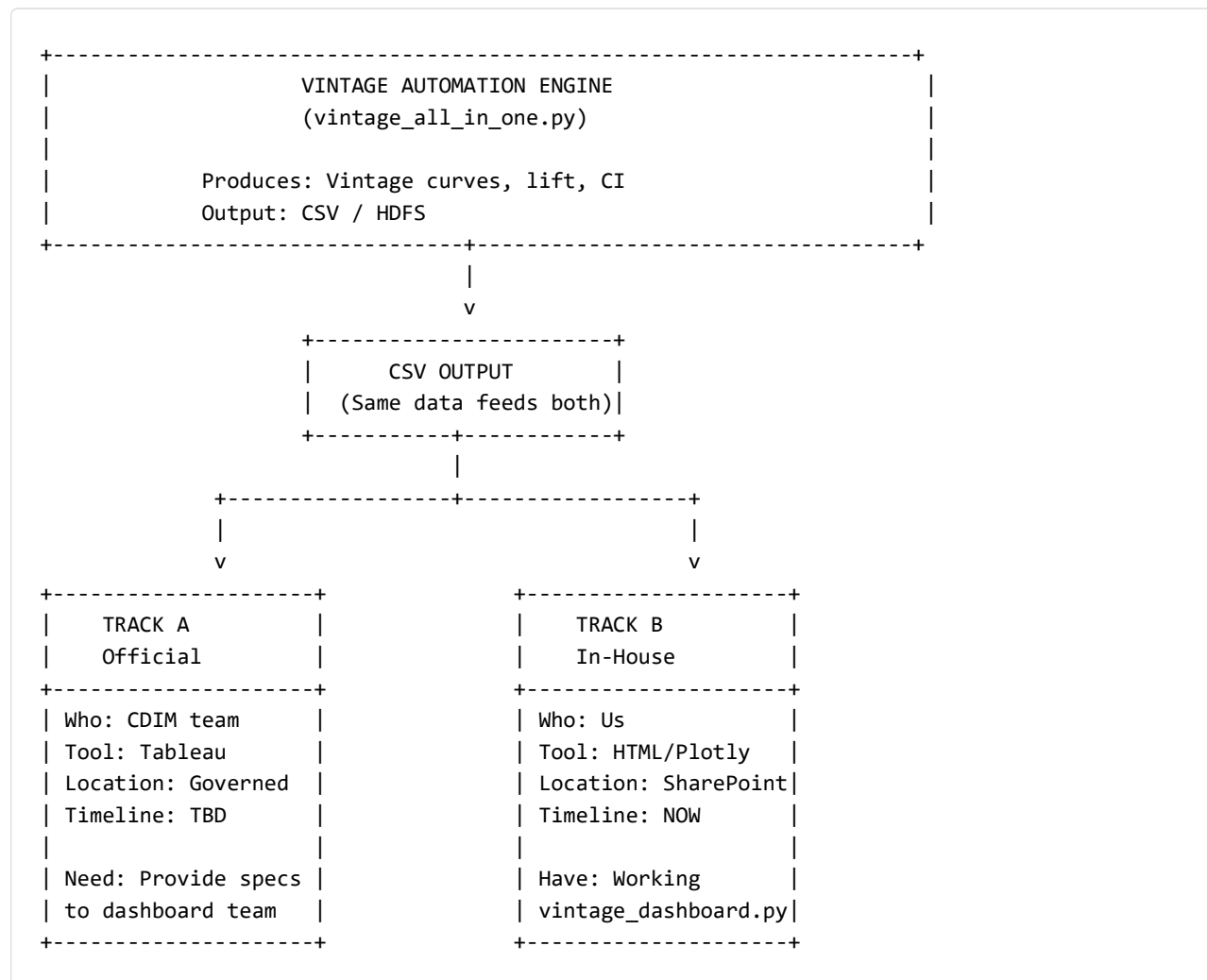
- X-axis: Days from treatment
- Y-axis: Cumulative rate (for selected metric)
- Lines: Test vs Control by cohort

Why This Matters

- See engagement **trends over time** per cohort
- Compare cohort performance on ANY metric
- Identify if recent cohorts have higher unsubscribe rates
- Director-level insight, not just analyst summary

Two Tracks for Dashboard (Running in Parallel)

We're pursuing **both tracks simultaneously** - they're not mutually exclusive.



Track	Purpose
Track A (Official)	For governance, official reporting, broad access
Track B (In-House)	For immediate needs, ad-hoc analysis, iteration

Next step for Track A: Meet with CDIM team to share specs and understand their timeline/capacity.

Track B status: Working - vintage_dashboard.py generates HTML with RBC colors, Plotly charts, campaign/cohort dropdowns.

Next Steps (Proposed)

Engine & Data (Both Tracks)

1. **Test current code** - Run VCN on actual Spark environment
2. **Validate results** - Compare against known vintage if available
3. **Get test group info** - Research TST_GRP_CD values for A/B test support
4. **Integrate channel breakdown** - Show in dashboard as additional detail

Track A: Official Dashboard

5. **Meet with CDIM team** - Share specs, understand timeline/capacity
6. **Provide data format specs** - What columns, what format they need
7. **Align on refresh cadence** - How often data updates

Track B: In-House Dashboard

8. **Finalize HTML dashboard** - vintage_dashboard.py ready
9. **Deploy to SharePoint** - Accessible for internal use
10. **Document for handoff** - Process for adding new campaigns

Key Message

Vintage Automation is not parallel work - it's **implementation of the SuperFact vision**:

- Pulls what we can from data NOW (experiment metadata, channel)
- Documents swap points for what will come from data LATER (campaign metadata, success library)
- Produces governed, consistent vintage curves with confidence intervals
- Proves the 4-layer architecture works in practice

This is a pilot. The patterns here scale to other campaign families beyond these 6.